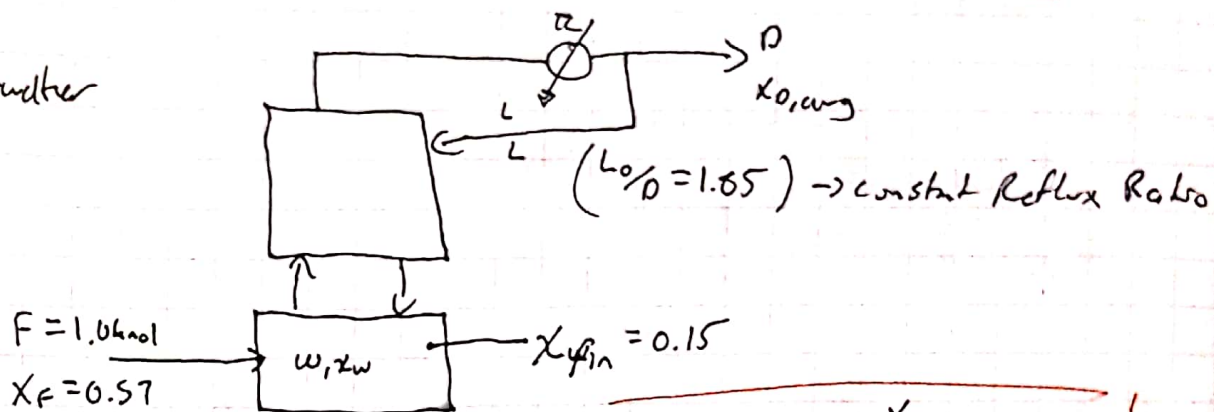


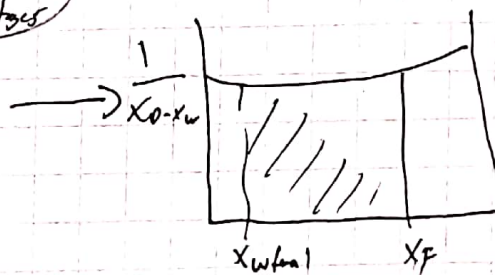
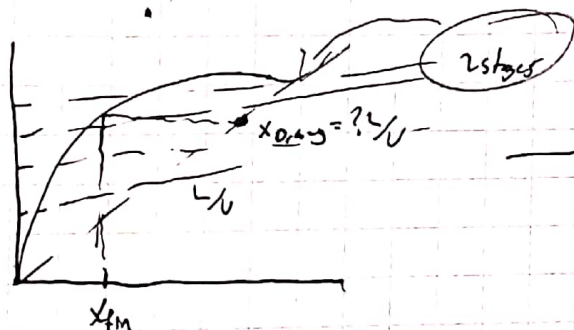
9.03

Methanol-water



Find W_{final} , D_{total} , $x_{D,avg}$
 $L/D \text{ constant}$

$$\text{Raleigh} \quad \ln \left(\frac{W_{final}}{F} \right) = - \int_{x_{final}}^{x_F} \frac{dx_W}{x_D - x_W}$$



$\rightarrow \text{Raleigh} \rightarrow W_{final}$
 $x_{D,avg}$
 D_{total}

Excel

$$OP \rightarrow y = \frac{L}{V} x + \left(1 - \frac{L}{V}\right) x_D$$

$$\frac{L}{V} = \frac{L/D}{1 + L/D}$$

then - Simpson's rule (excel)

$$W_{final} = F \exp(-A_{area}) = 0.38786$$

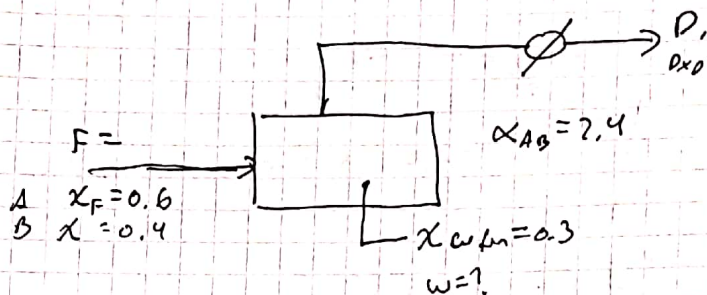
$$D_{total} = F - W_{final} = 0.61214$$

$$x_{D,avg} = \frac{F x_F - W_{final} x_{W,in}}{F - W_{final}} = 0.83611$$

Note: Don't have probe at here

0.6 Simple Batch $\alpha \cdot x_{0,avg} = 0.75, w=1, x_{w,fin}=0.3$

(2)



a) $F = 1.3 \text{ kmol}$ $x_{f,fin} = 0.3$ $w = 1$ $x_{0,avg} = ?$

$$W_{final} = F \exp \left(\frac{1}{\alpha-1} \ln \left(\frac{x_{fm}(1-x_F)}{x_F(1-x_{fm})} \right) + \ln \left(\frac{1-x_F}{1-x_{fm}} \right) \right)$$

$$= (1.3 \text{ kmol}) \exp \left(\frac{1}{2.4-1} \ln \left(\frac{0.3(1-0.6)}{0.6(1-0.3)} \right) + \ln \left(\frac{1-0.6}{1-0.3} \right) \right)$$

$$(W_{final} = 0.303586 \text{ kmol})$$

$$x_{0,avg} = \frac{F x_F - W_{final} x_{fm}}{F - W_{final}} = \frac{(1.3)(0.6) - (0.303586)(0.3)}{1.3 - 0.303586}$$

$$(x_{0,avg} = 0.691)$$

b) $F = 3.5 \text{ kmol}$

$\Rightarrow$ same eqn, same procedure

$$\Rightarrow W_{final} = F \exp(\dots)$$

$$= 3.5 \exp(\dots)$$

$$(W_{final} = 0.81735 \text{ kmol})$$

$$x_{0,avg} = \frac{(3.5)(0.6) - (0.81735)(0.3)}{3.5 - 0.81735}$$

$$(x_{0,avg} = 0.6914)$$

$C_1 F = 7.0 \text{ kmol} \times \{ x_{D, \text{avg}} = 0.75, \quad w = ? \text{ (1)} + x_{w, \text{fin}} = ? \text{ (2)} \}$

$\textcircled{1} \textcircled{2} = F \left(\frac{1}{\exp \alpha - 1} \ln \left(\frac{x_{f, \text{in}} (1 - x_{f, \text{r}})}{x_{f, \text{r}} (1 - x_{f, \text{in}})} \right) + \ln \left(\frac{1 - x_{f, \text{r}}}{1 - x_{f, \text{in}}} \right) \right) - w$

$\textcircled{2} x_{D, \text{avg}} = \frac{F x_{f, \text{r}} - w x_{f, \text{in}}}{F - w x_{f, \text{in}}}$

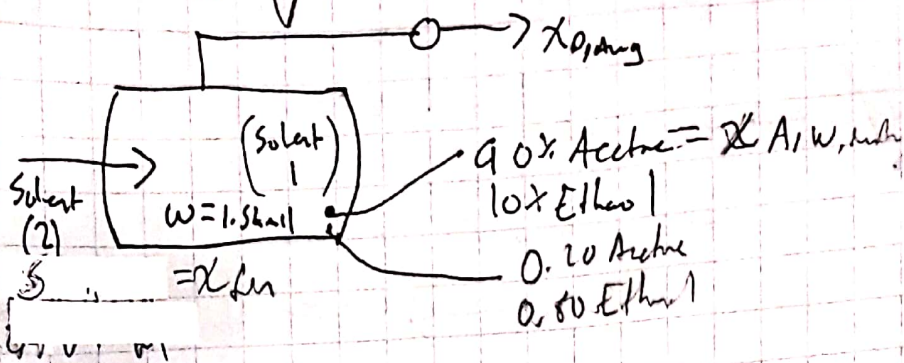
Unknowns $w = 1.211$

$x_{w, \text{fin}} = 0.502296$

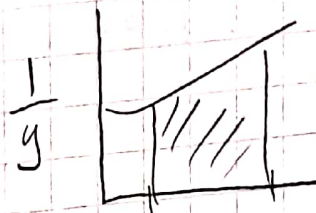
Excel solver



D.12 Switching Solvent Acetone & Ethanol



$$\frac{S}{W} = \int_{x_{oil}}^{x_{oil,initial}} \frac{dx_{oil}}{y}$$



$$\Rightarrow S = W \cdot A_{ren}$$

Break Exce!!

$$A_{ren} = 1.050395$$

$$S = (1.5)(1.050395)$$

$$[S = 1.575592] \text{ kmol} \checkmark$$

kmol Acetone evaporated?

$$[0.9 \rightarrow 0.2] \text{ Acetone}$$

$$\Rightarrow \Delta n_{up} = (x_{oil} - x_{oil,initial}) \cdot W$$

$$(0.9 - 0.2) \cdot 1.5$$

$$[A_{up} = 1.05] \text{ kmol} \checkmark$$

kmol ethanol evaporated?
0.1 $\rightarrow$ 0.8

$$[S = V]$$

$$S = V_A + V_E$$

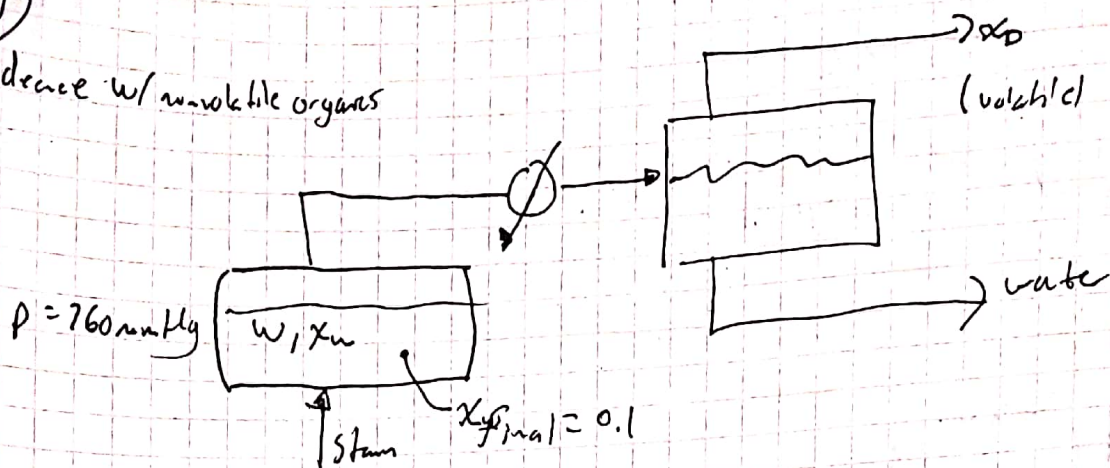
$$V_E = S - V_A = 1.57 - 1.05$$

$$[V_E = 0.525592] \text{ kmol} \checkmark$$

$$X_{oil,avg} = \frac{A_{up}}{S} = \frac{1.05}{1.57} = 0.667$$

D.19)

n-decane w/ nonvolatile organics



$$F = 16 \text{ kmol}$$

$$x_F = 0.9$$

a) At the final time, what is the temp in the still?

$$P = P_{org} + P_w = (VP_{org}) \cdot x_{org} + (VP_w) (x_w)^{P_{water}}$$

$$P = 760 \text{ mmHg} = (VP_{org}) \cdot (0.1) + (VP_w)$$

$$\text{Int } \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_0 - y_1}{x_0 - x_1}$$

$$y_0 = (x_0 - x_1) \left(\frac{y_2 - y_1}{x_2 - x_1} \right) + y_1$$

guess 1 $T = 99.5^\circ \text{C}$

$$(VP)_{org} = (99.5 - 95.5) \left(\frac{100 - 60}{108.6 - 95.5} \right) + 60 (VP)_w = 746.52 \text{ mmHg}$$

$$(VP)_{org} = 72.213 \text{ mmHg}$$

$$P = VP_w + VP_{org} x_{org}$$

$$= 746.52 + (72.213)(0.1)$$

$$P_1 = 753.741 \text{ mmHg}$$

guess 2 $T = 100^\circ \text{C}$

$$(VP)_{org} = (100 - 95.5) \left(\frac{100 - 60}{108.6 - 95.5} \right) + 60$$

$$(VP)_w = 760 \text{ mmHg}$$

$$P = 73.74 \text{ mmHg}$$

$$P = VP_w + VP(x)_{org}$$

$$(P = 767.374)$$

$$\Rightarrow T = 99.5 + \left(\frac{760 - 753.74}{767.3 - 753} \right) (100 - 99.5) \Rightarrow T = 99.74^\circ \text{C}$$

b) $w_{final} = ?$ $D_{total} = ?$

$$\ln\left(\frac{w_{final}}{F}\right) = - \int_{x_{Ln}}^{x_F} \frac{dx_w}{x_D - x_w} \quad x_D = 1$$

$$\Rightarrow w_{final} = F \exp + \left(\ln(1 - x_w) \right) \Bigg|_{x_{Ln}=0.1}^{x_F=0.9}$$

$$= 10 \exp (\ln(0.1) - \ln(0.9))$$

$$(w_{final} = 1.1111) \quad \checkmark$$

$$D_{tot} = F - w_{final} = 10 - 1.1111$$

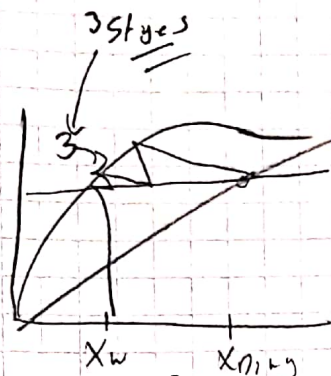
$$(D_{tot} = 8.8888) \quad \checkmark$$

$$c) \frac{n_w}{n_{org}} = \frac{P_{tot} - (V.P_{vcc}) x_{P_{tot} \text{ org}}}{x_P \lambda_{oc} \oplus x_{P_{tot} \text{ org}}}$$

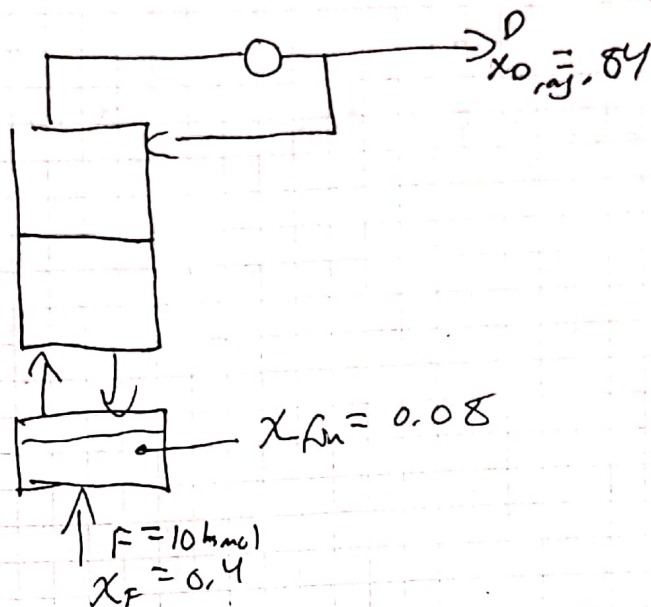
$$(V.P_{vcc}) = (99.74 - 95.5) \left(\frac{100 - 60}{108.6 - 95.5} \right) + 60 = 72.94$$

$$\left(\frac{n_w}{n_{org}} = \frac{260 - (72.94)(0.1)}{72.94 \oplus 0.1} = 103.145 \right) \quad \checkmark$$

$(\frac{L}{D})_{min} \times \rho_{constant}$



2



$x_{fm} = 0.08$

$F = 10 \text{ kmol/h}$
 $x_F = 0.4$

a) $L/D = ? \leftarrow x_F = 0.4 \quad L/V = \begin{matrix} 0.2 \\ 0.3 \leftarrow \\ 0.4 \\ 0.5 \end{matrix}$

$$y = (\frac{L}{D})x + (1 - \frac{L}{D})x_D$$

[?]

From Excel $L/V \approx 0.3$ ish

$$\Rightarrow (L/D = 0.32) \quad \checkmark$$

$$\Rightarrow \frac{L}{D} = \frac{L/V}{1 - L/V} = \frac{0.32}{1 - 0.32} = 0.4705 \quad \checkmark$$

b) $(L/D) = ? \leftarrow 0.08 \quad L/V = \begin{matrix} 0.6 \\ 0.7 \\ 0.8 \leftarrow \text{Hees number} \\ 0.9 \end{matrix}$

$$(L/V = 0.84)$$

$$\left(\frac{L}{D} = \frac{L/V}{1 - L/V} = \frac{0.84}{1 - 0.84} = 5.25 \right) \quad \checkmark$$

c) $D_{total} = ? \quad W_{final} = ?$
 $F = W + D \Rightarrow D = F - W$

$$F x_F = W x_W + D x_D$$

$$\Rightarrow W_{final} = \left(\frac{x_D - x_F}{x_D - x_W} \right) F = \left(\frac{0.84 - 0.4}{0.84 - 0.08} \right) 10 = 5.79 \text{ kmol} \quad \checkmark$$

$$\Rightarrow D = 10 - 5.79 = 4.21 \text{ kmol}$$

Equilibrium data

x	y	y=x	y(0) = OP	OP xD=.3	OP xD=.4	OP xD=.5	OP xD=.6	OP xD=.7	OP xD=.8	OP xD=.9
0	0	0	0	10.5263158	14.03508772	17.54385965	21.05263	24.5614	28.07018	31.57895
2	13.4	2	0.701754	11.8245614	15.33333333	18.84210526	22.35088	25.85965	29.36842	32.87719
4	23	4	1.403509	13.122807	16.63157895	20.14035088	23.64912	27.15789	30.66667	34.17544
6	30.4	6	2.105263	14.4210526	17.92982456	21.43859649	24.94737	28.45614	31.96491	35.47368
8	36.5	8	2.807018	15.7192982	19.22807018	22.73684211	26.24561	29.75439	33.26316	36.77193
10	41.8	10	3.508772	17.0175439	20.52631579	24.03508772	27.54386	31.05263	34.5614	38.07018
15	51.7	15	5.263158	20.2631579	23.77192982	27.28070175	30.78947	34.29825	37.80702	41.31579
20	57.9	20	7.017544	23.5087719	27.01754386	30.52631579	34.03509	37.54386	41.05263	44.5614
30	66.5	30	10.52632	30	33.50877193	37.01754386	40.52632	44.03509	47.54386	51.05263
40	72.9	40	14.03509	36.4912281	40	43.50877193	47.01754	50.52632	54.03509	57.54386
50	77.9	50	17.54386	42.9824561	46.49122807	50	53.50877	57.01754	60.52632	64.03509
60	82.5	60	21.05263	49.4736842	52.98245614	56.49122807	60	63.50877	67.01754	70.52632
70	87	70	24.5614	55.9649123	59.47368421	62.98245614	66.49123	70	73.50877	77.01754
80	91.5	80	28.07018	62.4561404	65.96491228	69.47368421	72.98246	76.49123	80	83.50877
90	95.8	90	31.57895	68.9473684	72.45614035	75.96491228	79.47368	82.98246	86.49123	90
95	97.9	95	33.33333	72.1929825	75.70175439	79.21052632	82.7193	86.22807	89.73684	93.24561
100	100	100	35.08772	75.4385965	78.94736842	82.45614035	85.96491	89.47368	92.98246	96.49123

L/D	1.85
L/V	0.649123
x	0
F	1

x_d-guess	x_w-2stages	1/(x_d - x/w)
0.3	0.02	3.571428571
0.4	0.03	2.702702703
0.5	0.04	2.173913043
0.6	0.07	1.886792453
0.7	0.12	1.724137931
0.8	0.25	1.818181818
0.9	0.57	3.03030303

x_w-2stages	1/(x_d - x/w)
0.15	1.7
0.36	2.2
0.57	3.03030303

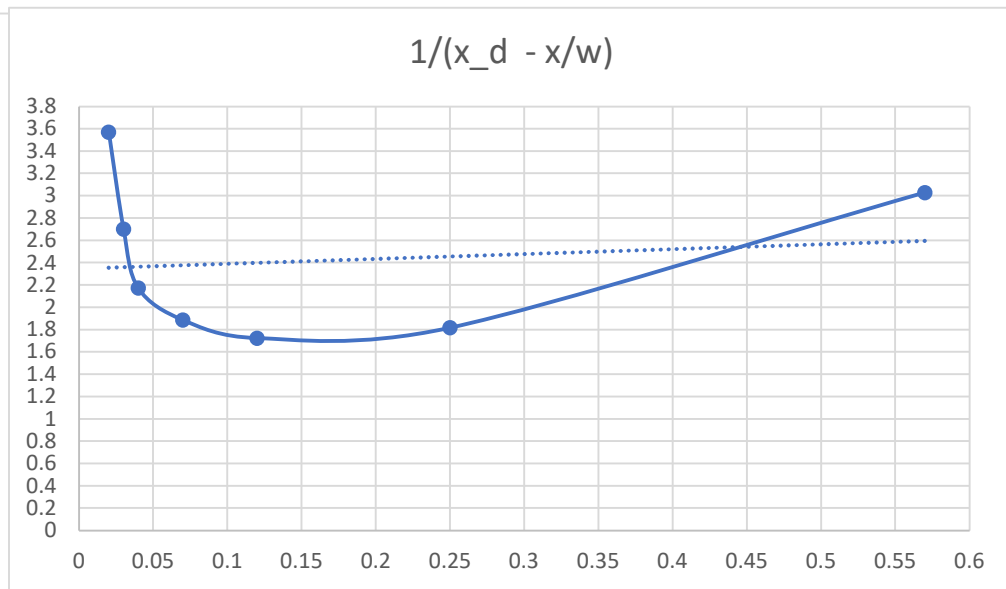
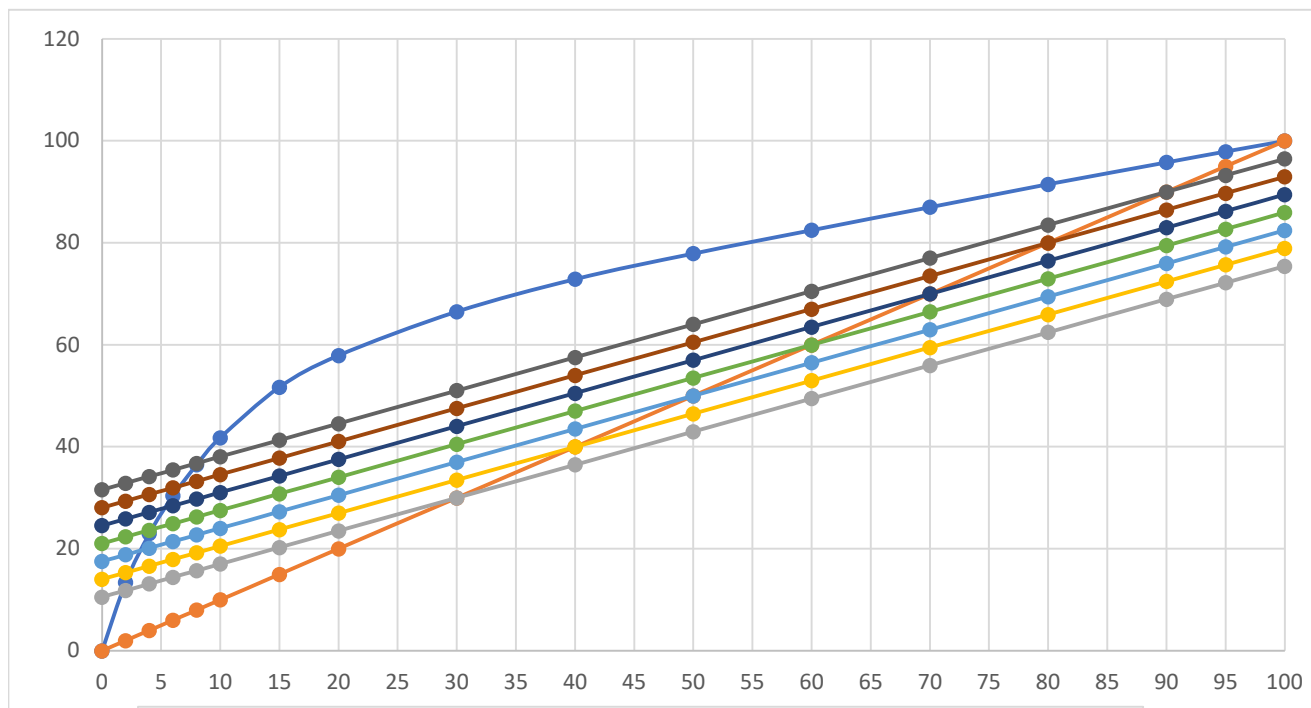
Simpsons A
0.947121

Wfinal
0.387856

D_tot
0.612144

x_d-avg
0.836113





F	2
x _F	0.6

alpha	2.4
D	
x _{d,avg}	0.75

W _{fin}	1.211121
x _{w,fin}	0.502296

	LHS	RHS
Eq1	0	-2.2E-09
Eq2	0.75	0.75

xA	yA	1/y	Area part
0	0		
0.1	0.262	3.816794	
0.15	0.348	2.873563	
0.2	0.417	2.398082	
0.25	0.478	2.09205	0.112253
0.3	0.524	1.908397	0.100011
0.35	0.566	1.766784	0.09188
0.4	0.605	1.652893	0.085492
0.5	0.674	1.48368	0.156829
0.6	0.739	1.35318	0.141843
0.7	0.8	1.25	0.130159
0.8	0.865	1.156069	0.120303
0.9	0.929	1.076426	0.111625
1	1	1	

Area(.9-.2)

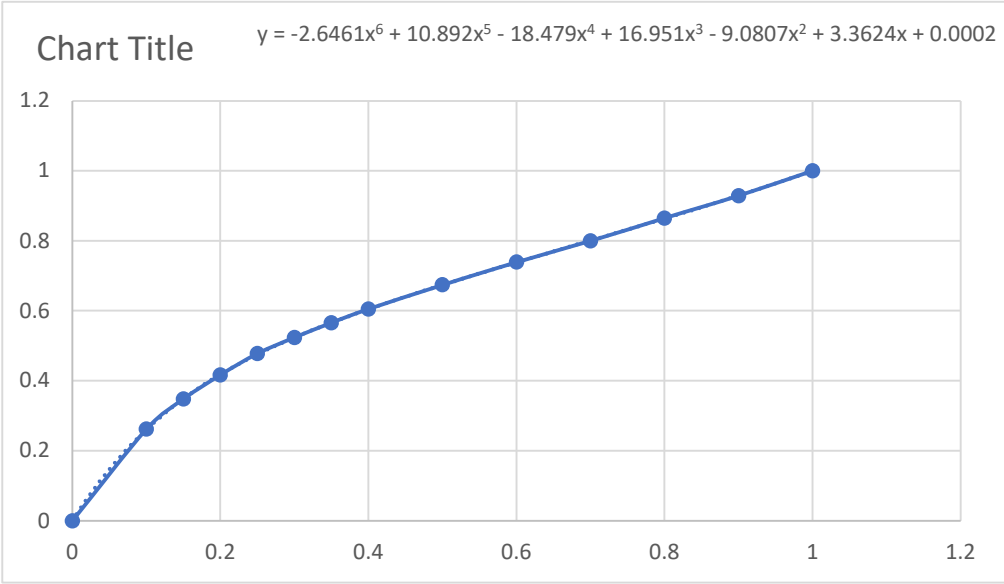
1.050395

W

1.5

S

1.575592



x	y	y=x
0	0	0
2	13.4	2
4	23	4
6	30.4	6
8	36.5	8
10	41.8	10
15	51.7	15
20	57.9	20
30	66.5	30
40	72.9	40
50	77.9	50
60	82.5	60
70	87	70
80	91.5	80
90	95.8	90
95	97.9	95
100	100	100

x_d,avg 84

L/V	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
67.2	58.8	50.4	42	33.6	25.2	16.8	8.4	
67.6	59.4	51.2	43	34.8	26.6	18.4	10.2	
68	60	52	44	36	28	20	12	
68.4	60.6	52.8	45	37.2	29.4	21.6	13.8	
68.8	61.2	53.6	46	38.4	30.8	23.2	15.6	
69.2	61.8	54.4	47	39.6	32.2	24.8	17.4	
70.2	63.3	56.4	49.5	42.6	35.7	28.8	21.9	
71.2	64.8	58.4	52	45.6	39.2	32.8	26.4	
73.2	67.8	62.4	57	51.6	46.2	40.8	35.4	
75.2	70.8	66.4	62	57.6	53.2	48.8	44.4	
77.2	73.8	70.4	67	63.6	60.2	56.8	53.4	
79.2	76.8	74.4	72	69.6	67.2	64.8	62.4	
81.2	79.8	78.4	77	75.6	74.2	72.8	71.4	
83.2	82.8	82.4	82	81.6	81.2	80.8	80.4	
85.2	85.8	86.4	87	87.6	88.2	88.8	89.4	
86.2	87.3	88.4	89.5	90.6	91.7	92.8	93.9	
87.2	88.8	90.4	92	93.6	95.2	96.8	98.4	

